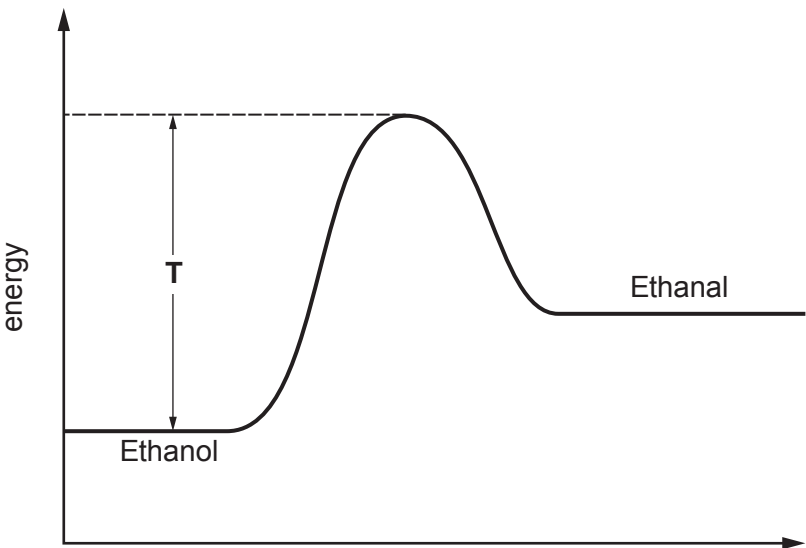


6. Alcohol dehydrogenase is an enzyme that catalyses the conversion of ethanol into ethanal.

The graph below shows the change in energy state when ethanol is converted to ethanal when no enzyme is present.



(a) (i) State what is represented by the region labelled **T**. [1]

(ii) Draw a curve on the diagram to show the change in energy level when alcohol dehydrogenase is present. [1]

(b) Describe how alcohol dehydrogenase can catalyse the conversion of ethanol into ethanal. [3]

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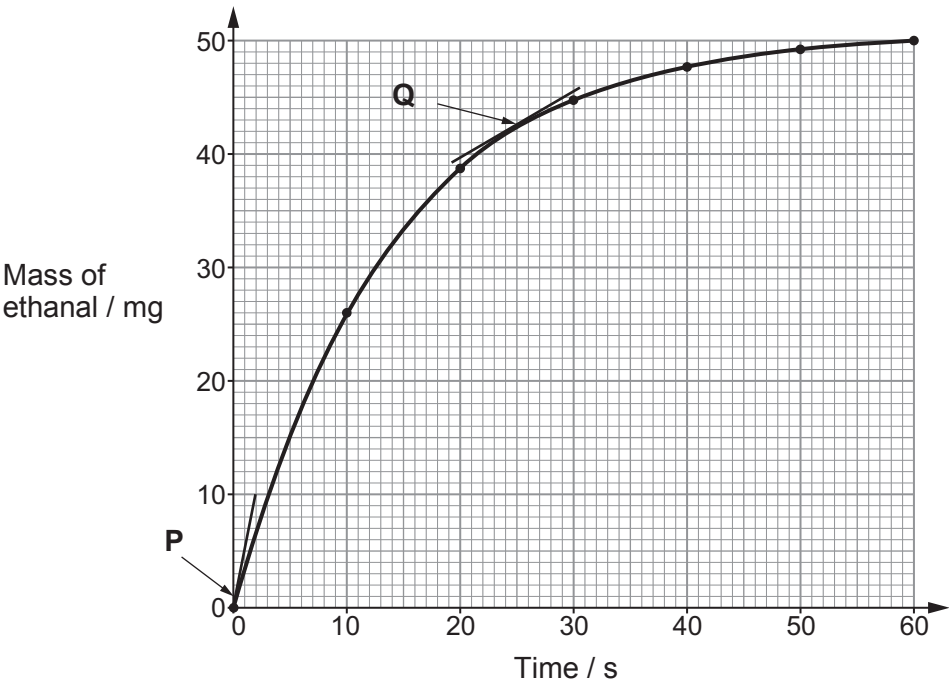
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- (c) A fixed mass of ethanol was added to a test tube containing alcohol dehydrogenase and a pH7 buffer solution. The test tube was incubated at 30°C and the mass of ethanal produced over time was recorded. The results are shown below.



- (i) The rate of reaction at **P** was  $5 \text{ mg s}^{-1}$ .  
Calculate the rate of reaction at **Q**. [2]

rate = .....  $\text{mg s}^{-1}$

- (ii) What conclusions can be drawn that would account for the difference in rate at **P** and at **Q**? [4]

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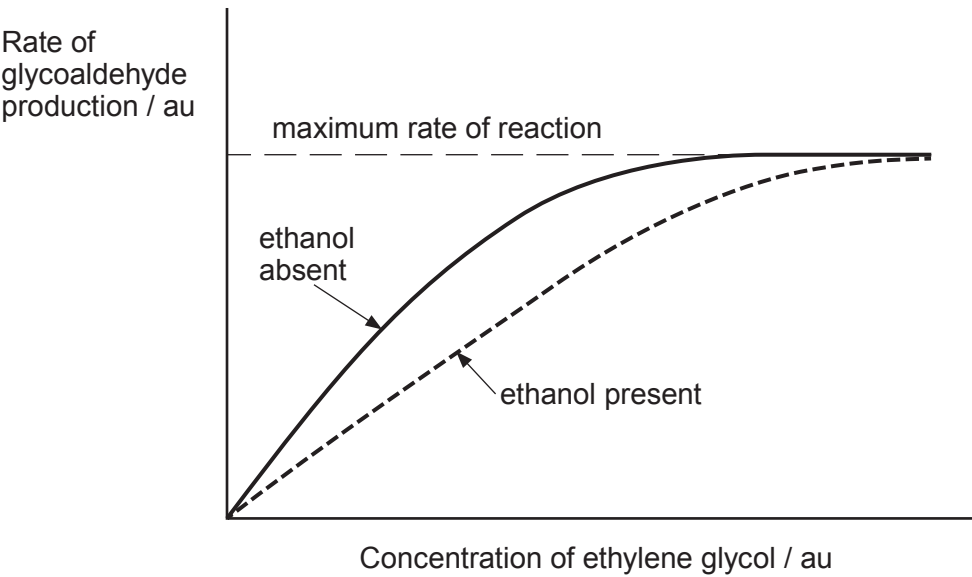


Examiner  
only

(d) Ethylene glycol is a colourless, odourless, sweet liquid, commonly found in antifreeze. It is highly toxic if ingested because once inside the body ethylene glycol is converted into glycoaldehyde. This reaction is also catalysed by alcohol dehydrogenase.

Treatment of ethylene glycol poisoning includes giving the patient ethanol, either intravenously or orally.

The graph below shows the rate of glycoaldehyde production in the presence, and absence of ethanol.



Use the information provided to explain why this treatment would reduce the effects of ethylene glycol poisoning. [5]

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